

Dr. Mahesh Shamrao Chaudhari

Signal Processing Engineer | Deputy Manager | PhD — IIT Patna

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PROFESSIONAL SUMMARY

PhD in Electrical Engineering from IIT Patna (CGPA 9.5/10) with deep expertise in RF signal processing, OFDM/OFDMA synchronization, deep learning for wireless communication, and SDR-based RF sensing. Currently serving as Deputy Manager – Signal Processing at Paras Anti-Drone Technologies Pvt. Ltd., leading the SkySense project with a team of 6–8 engineers to deliver passive RF-based drone detection and CNN/LSTM-based drone classification systems. Actively contributing to satellite communication system development. Published in IEEE Communications Letters and IEEE Transactions on Instrumentation & Measurement. IEEE Member.

CORE COMPETENCIES

Signal Processing: OFDM/OFDMA, CFO/STO Estimation, Passive RF Sensing, Spectrum Analysis, Synchronization

AI / Machine Learning: Deep Learning, CNN, LSTM, Attention Mechanisms, RF Fingerprinting, Signal Classification

Wireless & Standards: 5G NR (3GPP Rel-15), SC-FDMA, D2D Communication, Satellite Communication, SDR

Programming: Python, MATLAB, C, Embedded C, Java

Frameworks & Tools: TensorFlow, Keras, LabVIEW, MATLAB/Simulink, Proteus, Xilinx, MPLAB-X, NS-2/NS-3

Hardware Platforms: USRP B210, USRP-2920, NI RF Testbed, Pluto SDR, RTL SDR

Domains: Anti-Drone Systems, UAV Detection, Passive RF Intelligence, Satellite Communication

PROFESSIONAL EXPERIENCE

Deputy Manager – Signal Processing

Dec 2025 – Present

Paras Anti-Drone Technologies Pvt. Ltd. (Subsidiary of Paras Defence and Space Technologies Ltd.), Navi Mumbai

- ▶ Led the SkySense anti-drone project, managing a cross-functional team of 6–8 engineers across RF, deep learning, and embedded domains.
- ▶ Architected a passive RF listening pipeline for real-time UAV detection using signal intelligence techniques and SDR-based signal acquisition.
- ▶ Designed and trained CNN/LSTM-based deep learning models for drone classification using RF fingerprinting, achieving 95% classification accuracy on field-collected RF data.
- ▶ Oversaw SDR-based spectrum analysis, RF signal preprocessing, and feature extraction workflows for production deployment.
- ▶ Currently developing advanced signal processing algorithms and communication modules for a satellite communication project.

Assistant Professor

Aug 2024 – Dec 2025

Thapar Institute of Engineering and Technology (TIET), Patiala

- ▶ Delivered four courses: Digital Signal Processing (UEC502), Embedded Systems (PVL202), Microcontroller & Embedded Systems (UEC702), and Electrical & Electronics Engineering (UEC013).
- ▶ Designed lab-integrated curricula aligned with AICTE guidelines; mentored student projects in signal processing and embedded systems.

Senior Research Fellow – Deep Learning for Wireless Communication

May 2023 – Dec 2023

Indian Institute of Science (IISc), Bangalore

- ▶ Investigated deep learning-based synchronization and parameter estimation algorithms for OFDM systems under frequency-selective fading channels.
- ▶ Implemented and validated DL models on NI RF testbeds under real-world channel conditions; results contributed to IEEE journal publications.

Manager – Test & Development

Dec 2023 – Jun 2024

Anvin Electronics Pvt. Ltd.

- ▶ Managed test and development workflows for RF and embedded electronics systems; developed automated testing frameworks to accelerate hardware verification.

Student Trainee (Internship)

Aug 2022 – Feb 2023

Samsung R&D India, Bangalore

- ▶ Worked on 5G NR physical layer signal processing research; prototyped and validated algorithms in MATLAB and Python.

Research Assistant (Internship)

Jul 2019 – Nov 2019

Nanyang Technological University (NTU), Singapore

- ▶ Implemented a radar system using Software Defined Radios (SDRs), covering signal transmission, echo reception, and target detection using USRP platforms and MATLAB-based signal processing.

EDUCATION

PhD — Electrical Engineering | IIT Patna

2023 | CGPA: 9.5/10

Thesis: Estimation and Synchronization for Single Carrier and OFDM Systems Using Neural Network

Supervisor: Dr. Sudhan Majhi, Associate Professor, IISc Bangalore

ME — Electronics & Communication Engineering (Wireless Communication) | BIT Mesra

2017 | 91.6%

BE — Electronics & Telecommunication Engineering | PVG's COET, University of Pune

2014 | 62.94%

SELECTED PUBLICATIONS (IEEE JOURNALS)

1. M. S. Chaudhari, S. Majhi, and S. Jain, "CNN-Attention-DNN Design for CFO Estimation of Non-Pilot-Assisted OFDM System," IEEE Communications Letters, vol. 27, no. 2, pp. 551–555, Feb. 2023.
2. M. S. Chaudhari and S. Majhi, "STO Estimation for OFDM System Using CDM," IEEE Communications Letters, vol. 26, no. 11, pp. 2651–2655, Nov. 2022.
3. M. S. Chaudhari et al., "Design and Testbed Implementation of Blind Parameter Estimated OFDM Receiver," IEEE Transactions on Instrumentation and Measurement, vol. 71, Art. no. 5501211, 2022. DOI: 10.1109/TIM.2021.3124833

PROFESSIONAL ACTIVITIES & MEMBERSHIPS

- ▶ IEEE Member (No. 97339143) — Communication Society
- ▶ TPC Member: International Conference on 6G Communications, Networking & Signal Processing (SGCNSP), NTU Singapore, Dec 2023.
- ▶ Invited Talk: ICMNWC-2023, Sri Siddhartha Institute of Technology, Tumkuru (IEEE Bangalore), Dec 2023.
- ▶ Training Conducted: Wireless Algorithms on Pluto SDR & RTL SDR — Ballari Institute of Technology, Aug 2023.
- ▶ Peer Reviewer: IEEE Wireless Comm. Letters, IEEE Comm. Letters, IEEE Trans. Vehicular Technology, IEEE Trans. Wireless Comm., IEEE IoT Journal, IEEE Signal Processing Letters.

CERTIFICATIONS & NOTABLE TRAINING

- ▶ LabVIEW Core 1, Core 2, NXG Core 1 — National Instruments (Certified)
- ▶ GATE Qualified — 2015, 2016, 2017
- ▶ Short-Term Course: 4G/5G MIMO-OFDM Design using MATLAB/Python/Simulink — IIT Kanpur, Jun 2022
- ▶ Workshop: MIMO-OFDM Wireless Communications — IIT Kanpur, Feb 2016

PERSONAL DETAILS

Date of Birth: 17 February 1993

Languages: English, Hindi, Marathi

Location: Ahmednagar, Maharashtra, India (Currently: Navi Mumbai)